Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-II Physics Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC-II) Physics. It contains comments on candidates' responses to the 2017 SSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

In questions that required short written answers, students had knowledge of the correct answer, but were undisciplined in communicating that knowledge. This was particularly evident in Question 5 and 7. Students who considered how to communicate their knowledge in all types of questions were generally much more successful. Communication includes drawing field lines and vectors, labelling diagrams, using exact terminology, using correct formulae and answering the specific question asked.

This year, candidates performed well on questions related to the simple harmonic motion and waves, current electricity, electromagnetism, electronics and radioactivity. Whereas low-scoring candidates did not perform well in questions based on making and completing ray diagrams of the transparent materials, word problems based on the SHM, Coulombs law and refractive indices and also had problem in correctly articulating their answers on the basic definition, reasoning and discussion type questions.

Detailed Comments:

Constructed Response Questions (CRQs)

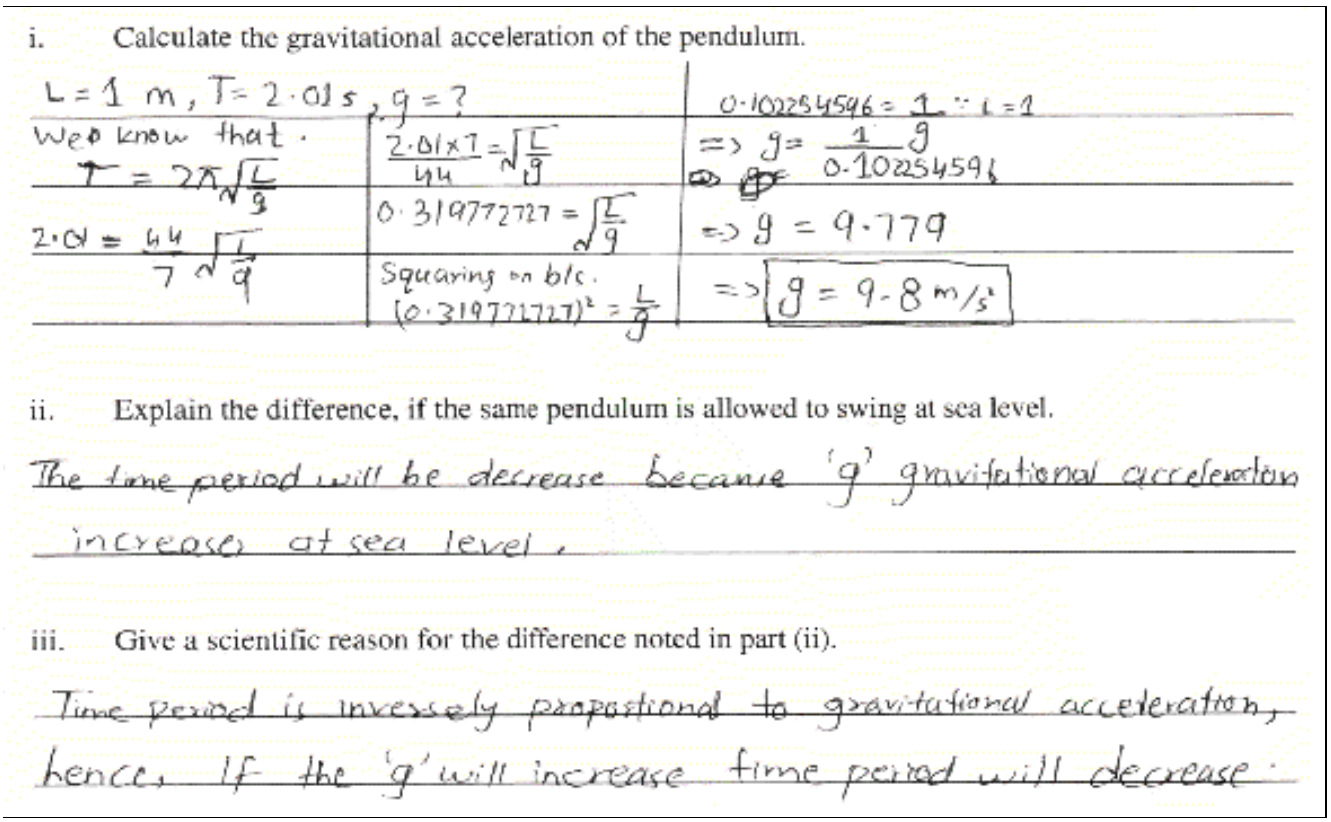
Question 1:

A pendulum of length 1.00 m has a period of 2.01 s, when it is allowed to swing at a height of 8954.0 m on Mount Everest.

i. Calculate the gravitational acceleration of the pendulum.
ii. Explain the difference, if the same pendulum is allowed to swing at sea level.
iii. Give a scientific reason for the difference noted in part (ii).

Better responses mentioned the correct formula of gravitational acceleration, $T = 2\pi\sqrt{\frac{l}{g}}$ or $g = \frac{4\pi^2 l}{T^2}$ and calculated the value, $g = 9.77\,m/s^2$. In the second part of the question, high-scoring candidates explained that at the sea level, a pendulum will swing with a slightly different period than at the top of Mount Everest and the clock's time at the sea level will show less time of oscillations. Some candidates wrote about decrease in time period and in the last part, candidates wrote the correct scientific reason that acceleration due to gravity (g) varies with height and length: $g \;\alpha\; \frac{1}{T}$.

Example:



i. Calculate the gravitational acceleration of the pendulum.

L = 1 m, T = 2.01 s, g = ?

We know that: $T = 2\pi\sqrt{\frac{L}{g}}$

$2.01 = \frac{44}{7}\sqrt{\frac{1}{g}}$

$\frac{2.01 \times 7}{44} = \sqrt{\frac{1}{g}}$

$0.319772727 = \sqrt{\frac{1}{g}}$

Squaring on b/c. $(0.319772727)^2 = \frac{1}{g}$

$0.102254596 = \frac{1}{g}$ ∵ 1 = 1

$\Rightarrow g = \frac{1}{0.102254596}$

$\Rightarrow g = 9.779$

$\Rightarrow$ **g = 9.8 m/s²**

ii. Explain the difference, if the same pendulum is allowed to swing at sea level.

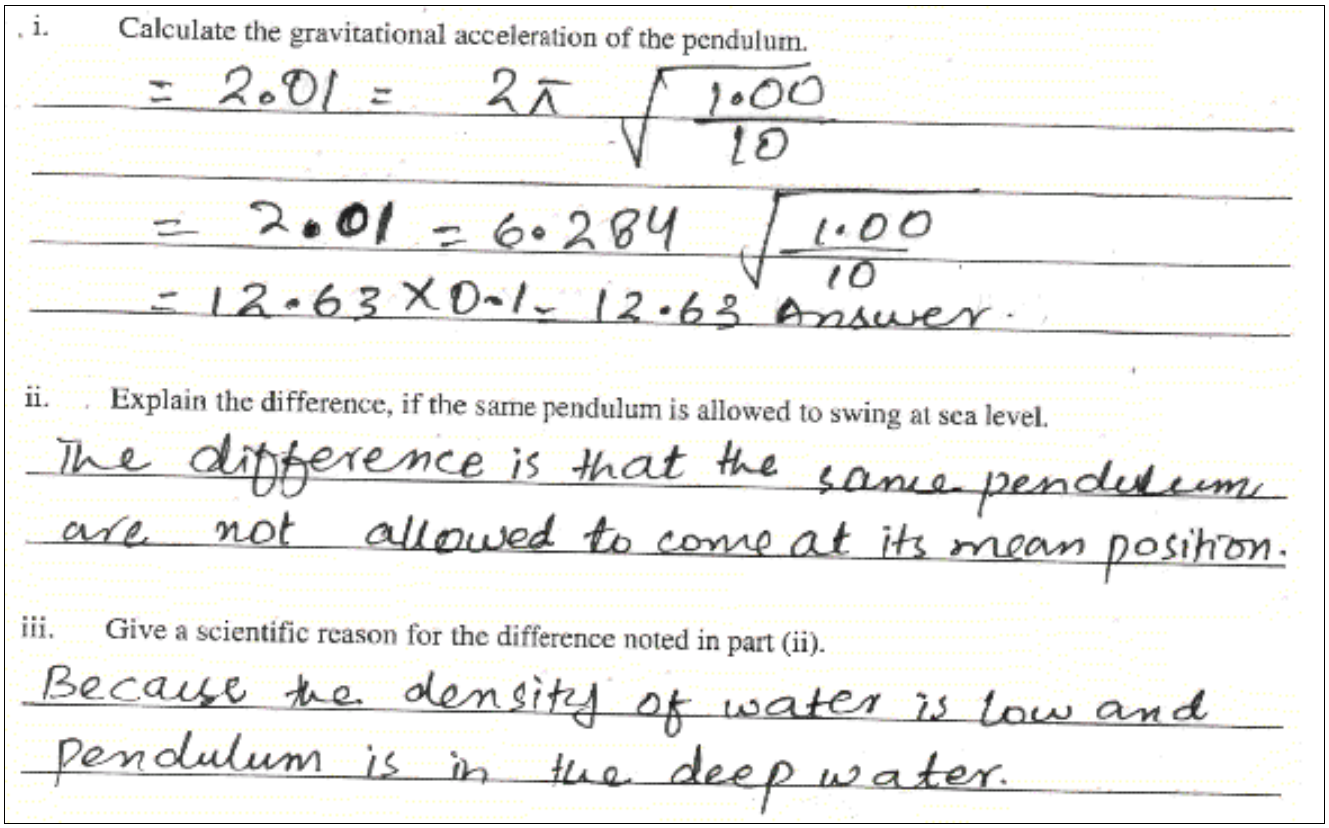
The time period will be decrease became 'g' gravitational acceleration increase at sea level.

iii. Give a scientific reason for the difference noted in part (ii).

Time period is invessely proportional to gravitational acceleration, hence, If the 'g' will increase time period will decrease.

Weaker responses showed that candidates lacked understanding of handling and calculating word problems, some candidates wrote the correct formula of time period but were unable to calculate the correct value. In the second part of the question, candidates wrote that 'difference in the time period is only because of the mean position or atmospheric conditions'; this was completely irrelevant to the demand of the question. While giving reason in the last part, low-scoring candidates wrote about the physical quantitates, i.e. density, mass, pressure and height of the pendulum placed at different positions. Some candidates also wrote that there is no difference in the time period of a simple pendulum when it is placed at sea level or at some height.

Example:



i. Calculate the gravitational acceleration of the pendulum.

$= 2.01 = 2\pi\sqrt{\frac{1.00}{10}}$

$= 2.01 = 6.284\sqrt{\frac{1.00}{10}}$

$= 12.63 \times 0.1 = 12.63$ Answer.

ii. Explain the difference, if the same pendulum is allowed to swing at sea level.

The difference is that the same pendulum are not allowed to come at its mean position.

iii. Give a scientific reason for the difference noted in part (ii).

Because the density of water is low and pendulum is in the deep water.

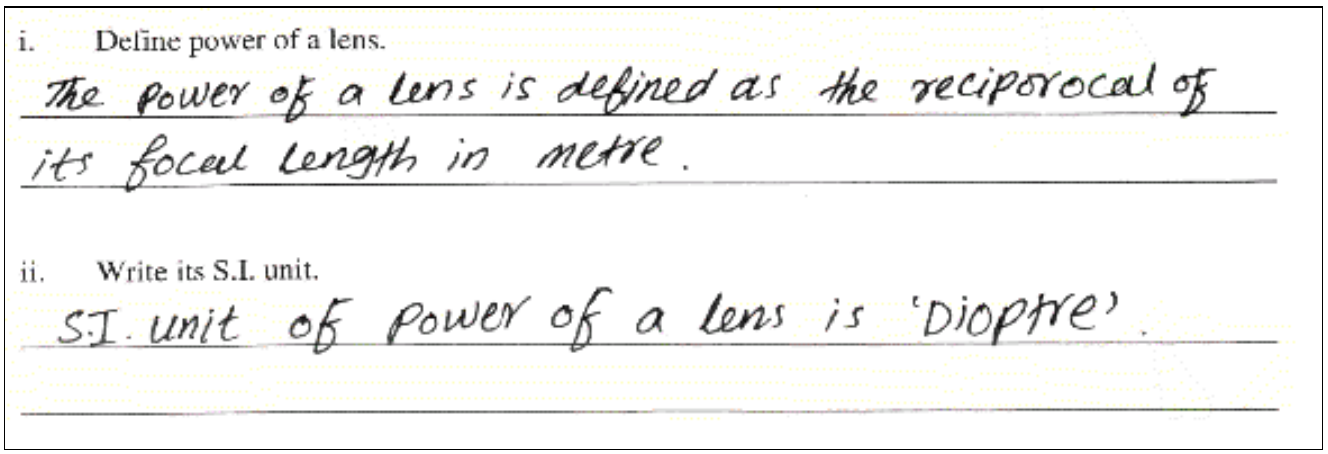
Question 2:

i. Define power of a lens?

ii. Write its S.I. unit.

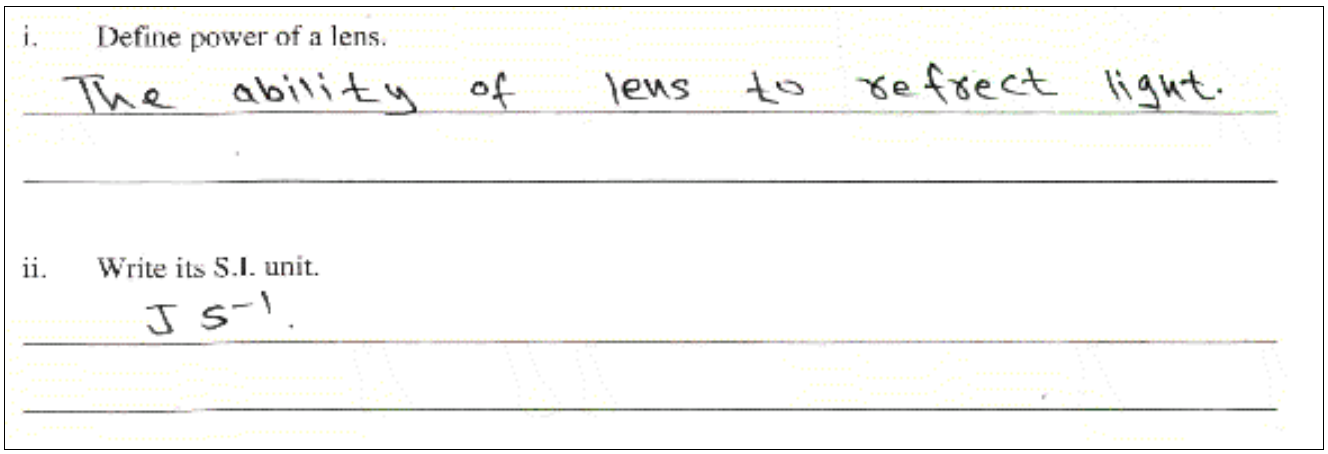
Better responses correctly defined the power of a lens, i.e. reciprocal of the focal length in metre and mentioned its correct S.I. unit, i.e. diopter. Some of the high-scoring candidates also wrote the formula of power as, $P = \frac{1}{\textit{focal length in metres}}$.

Example:



Weaker responses revealed that candidates were unable to write the proper definition of power and missed out the important part of definition, i.e. 'metre'. Low-scoring candidates wrongly wrote joules, joules per second and newton as the S.I. unit of power.

Example:



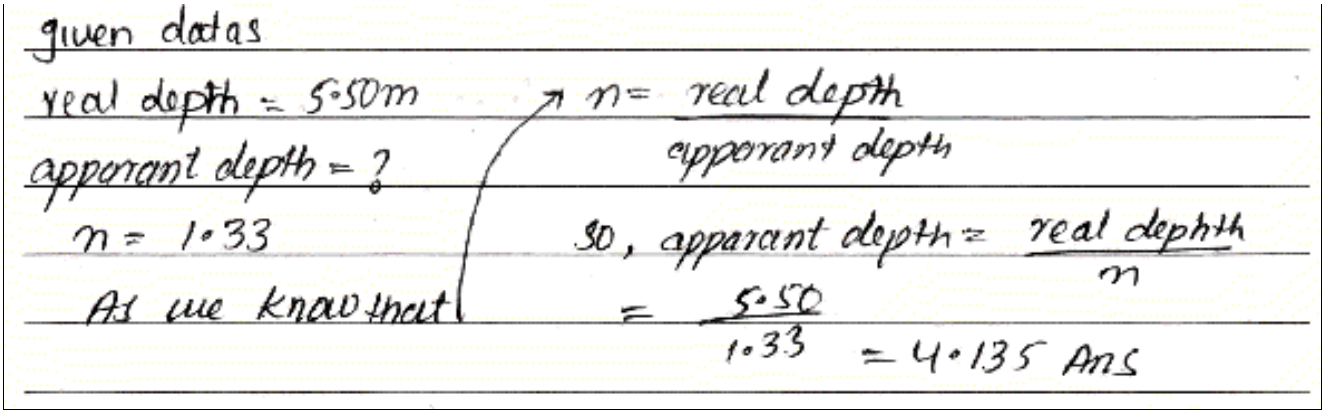
Question 3:

If the real depth of a swimming pool is 5.50 m, then calculate the apparent depth of the swimming pool.

(**Note**: The refractive index *n* of water is 1.33)

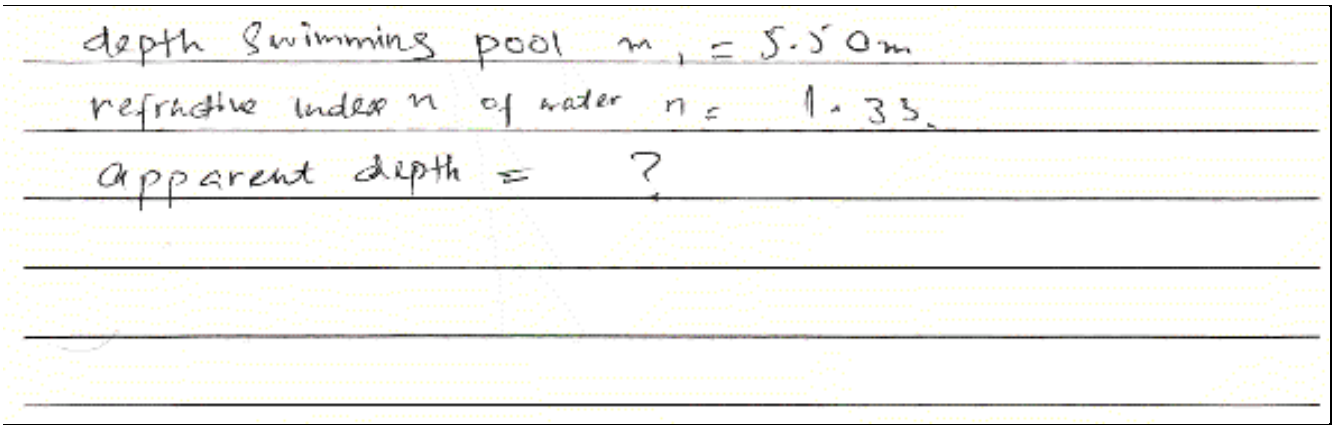
Better responses correctly calculated the apparent depth of the swimming pool by putting the correct formula, $n = \frac{\text{Real depth}}{\text{Apparent depth}}$, correct substitution and obtained the correct answer, 4.13 m.

Example:



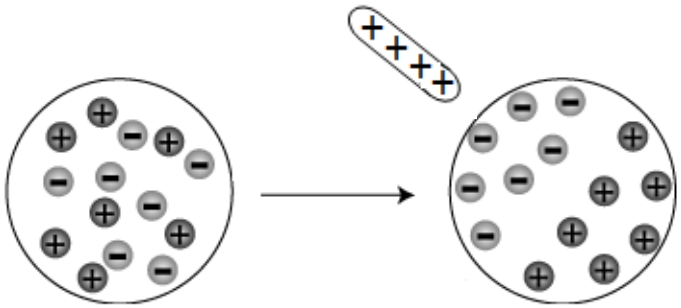
Weaker responses showed candidates' inability to calculate the correct answer. Some of the low-scoring candidates extracted the correct data, however they wrote incorrect formula of the refractive index, i.e. $n = \frac{\sin \angle r}{\sin \angle i}$.

Example:



Question 4:

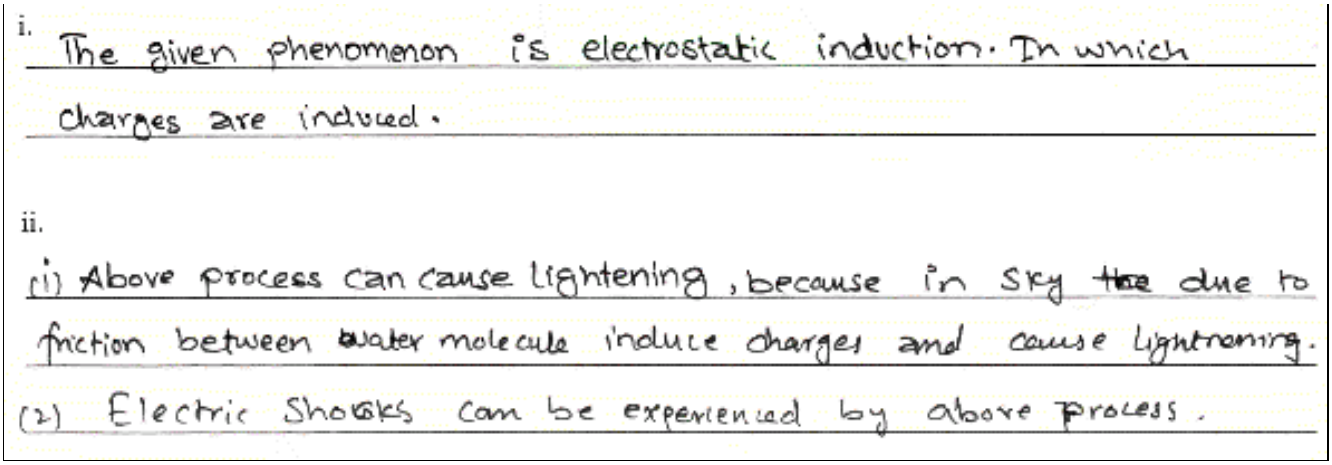
i. Identify the given phenomenon in the given diagram.



ii. Mention any TWO hazards of the phenomenon identified in part (i).

Better responses showed that candidates were able to correctly identify the phenomenon, i.e. electrostatic induction. High scoring candidates correctly mentioned more than two hazards of electrostatic induction, e.g. lightening, production of charges during fueling vehicles, oil tankers and aircrafts, touching something with a large electric charge on it will flow electric charge through the body of a person and causes an electric shock.

Example:


i. The given phenomenon is electrostatic induction. In which charges are induced.

ii. (1) Above process can cause lightening, because in sky due to friction between water molecule induce charges and cause Lightening.

(2) Electric Shocks can be experienced by above process.

Weaker responses provided incorrect identification of the phenomenon, e.g. candidates mentioned mutual induction, radioactivity, hydrogenation and electromagnetic induction. In the second part of the question, low scoring candidates mentioned attraction or repulsion of charges, energy loss or gain, nuclear power, explosion of atomic bomb, ratio between input and output on work done, which were not the demand of the question.

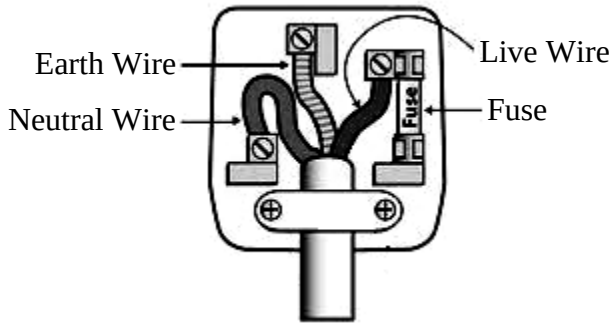
Example:

i. Electromagnetic induction.

ii.
↳ Energy is lose in this process.
↳ Output is less than input.

Question 5:

The given figure shows a power plug.



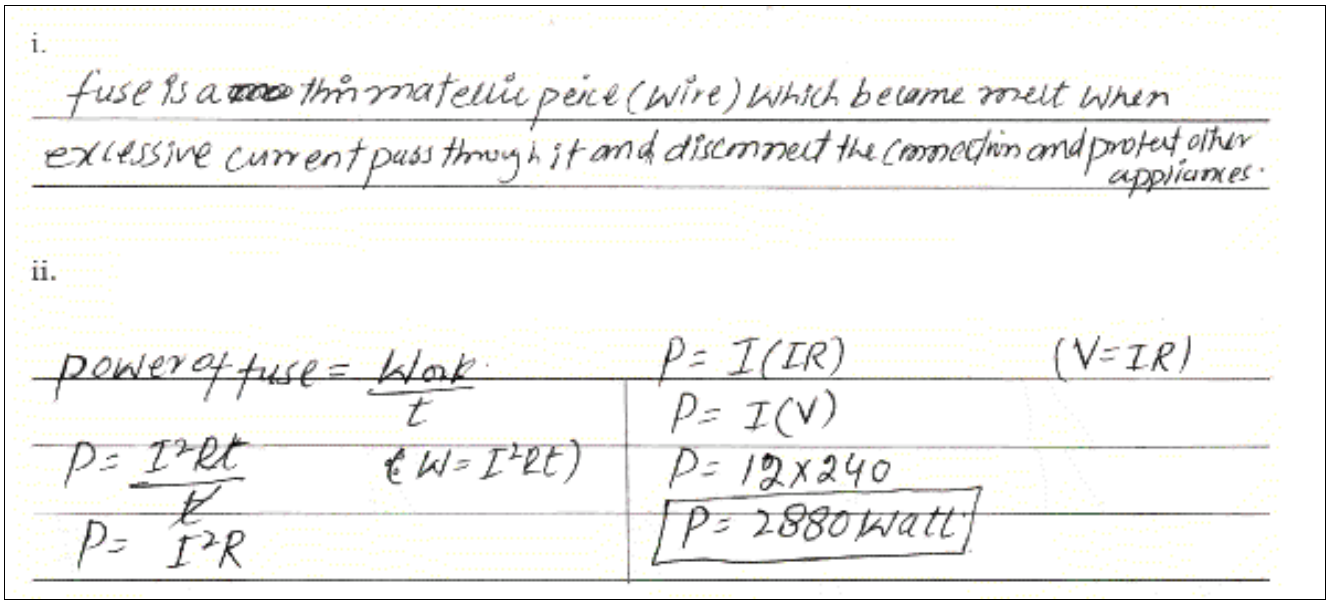
i. How is the safety of an electrical appliance secured by using a fuse in the power plug?

ii. If the rating of the fuse shown in the given figure is 12 ampere and it is working at 240 volt, then calculate the power of the fuse.

The question was challenging for most of the candidates.

Better responses showed that candidates correctly wrote about the safety of an electrical appliance secured by using a fuse. Most candidates mentioned that 'when excess of electric current flows through the fuse, it will heat up and burn and break the circuit and stop the flow of electric current'. In the second part of the question, high scoring candidates correctly applied the formula of power, i.e. P = V I and obtained the correct value of the power of the fuse, i.e. P = 2880 W.

Example:


i.

fuse is a thin metallic peice (wire) which become melt when excessive current pass through it and disconnect the connection and protect other appliances.

ii.

power of fuse = Work / t

$P = \frac{I^2Rt}{t}$ (∵ $W = I^2Rt$)

$P = I^2R$

$P = I(IR)$ ($V = IR$)

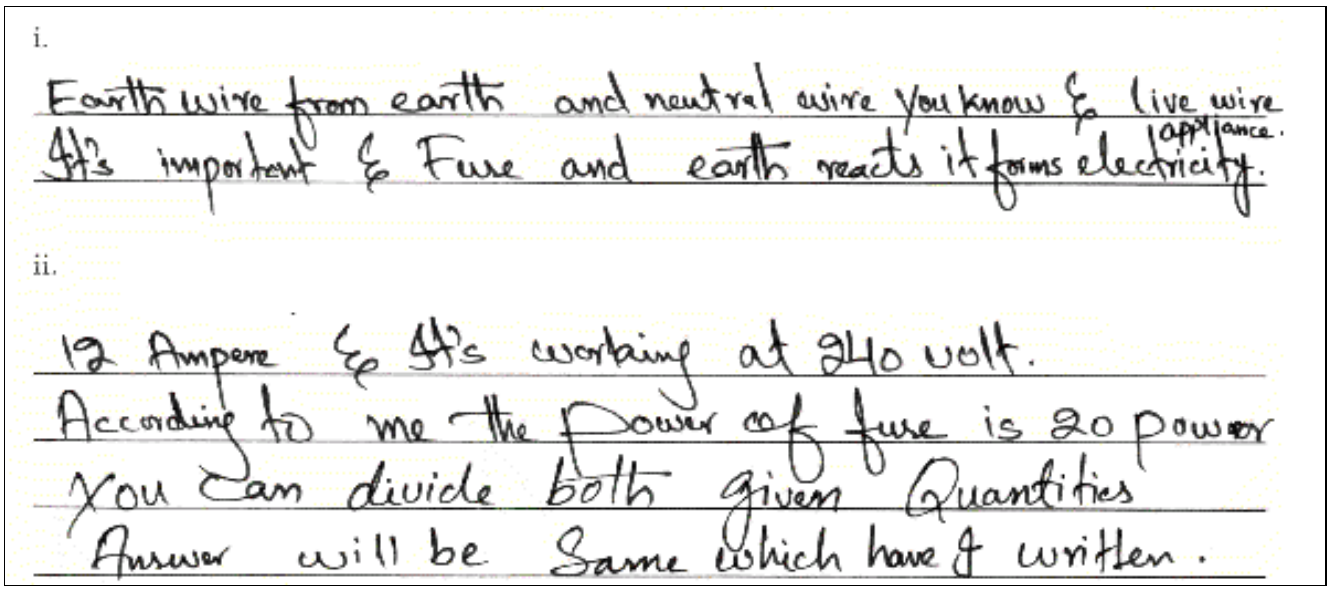
$P = I(V)$

$P = 12 \times 240$

$P = 2880$ Watt

Weaker responses reflected that candidates lacked knowledge about live, neutral and earth wires, the connection of fuse with power plug or live wire. Most candidates were unable to discuss melting or burning of fuse. Low scoring candidates used wrong formula of power, i.e. $P = \frac{V}{A}$ or $P = \frac{V}{I}$ or $W = \frac{Q}{V}$ and lost marks. Few of the candidates did not show mathematical calculation; instead, they explained this part in words.

Example:


i.

Earth wire from earth and neutral wire you know & live wire it's important & Fuse and earth reacts it forms electricity. appliance.

ii.

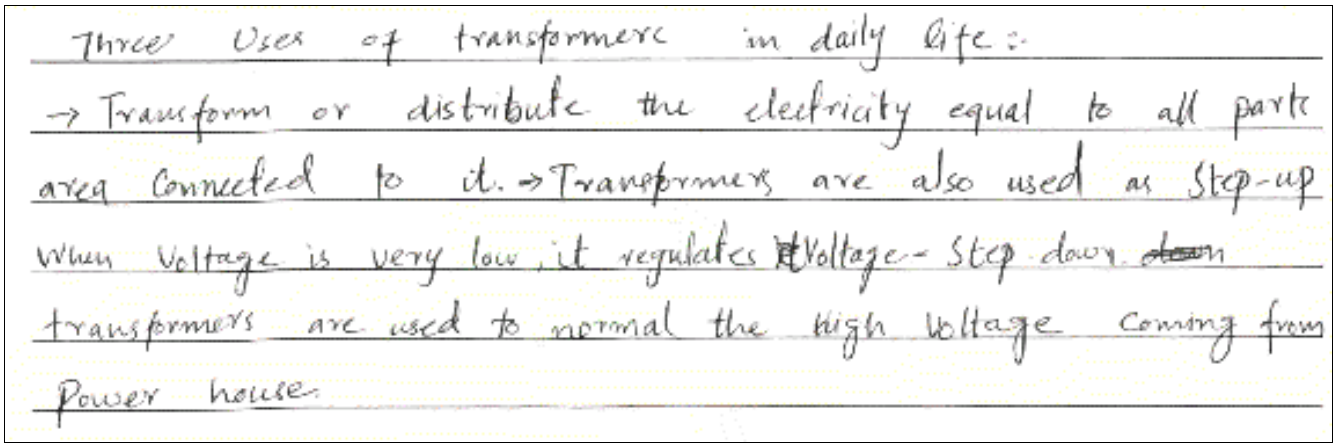
12 Ampere & it's working at 240 volt.
According to me the power of fuse is 20 power
You can divide both given Quantities
Answer will be Same which have & written.

Question 6:

Write THREE uses of transformers in our electric power system.

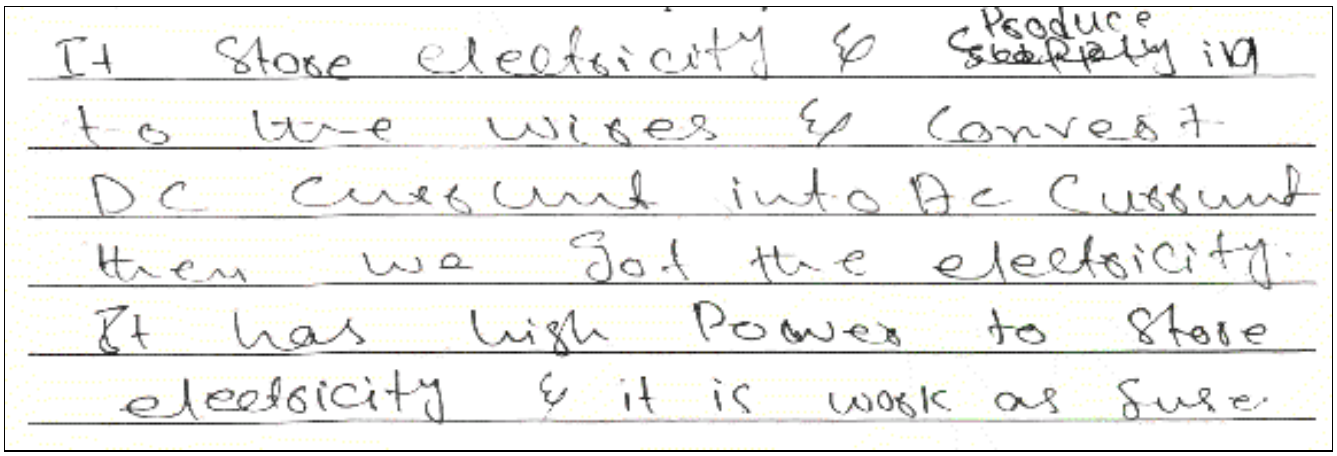
Better responses correctly wrote three uses of transformers. Most successful candidates mentioned any three of the following uses, e.g. used as step-up or step-down transformer, distribute or transfer electricity to far off places, reduce power loss, maintain current in the circuit, used in electric circuits to change the voltage and convert the alternating current into direct current.

Example:


Three Uses of transformers in daily life :-
→ Transform or distribute the electricity equal to all parts area Connected to it. → Transformers are also used as Step-up When Voltage is very low, it regulates Voltage - Step down ~~down~~ transformers are used to normal the high voltage coming from Power house.

Weaker responses wrote irrelevant uses of transformers, e.g. transformers are used to store electric current, used in domestic purposes in washing machines, juicers, and ovens or in X-rays machines in hospitals.

Example:


It Store electricity & Produce ~~Store~~ Reply in to the wires & Convert DC Current into Ac Current then we Got the electricity. It has high Power to Store electricity & it is work as fuse.

Question 7:

A truth table of a logic gate is given below.

A	B	$X = \overline{A.B}$
0	0	1
1	0	1
0	1	1
1	1	0

i. Identify the given logic gate.
ii. Draw the symbol of the logic gate.
iii. Write its mathematical operation.

The question was generally not well-attempted. Hence, it reflects that detailed attention is needed to explore this topic in the classrooms. A number of errors were displayed in this question, few common errors have been discussed below in the 'weaker responses' section.

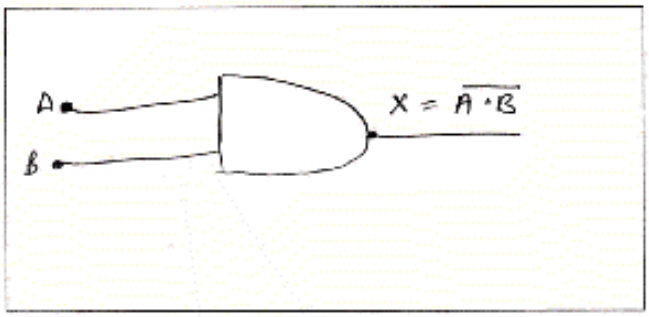
Better responses correctly identified the logic gate, i.e. NAND gate. High scoring candidates correctly drew its symbol and wrote the correct mathematical operation, formula, $X = \overline{A.B}$.

Example:

i. Identify the given logic gate.

The given gate is called NAND gate.

ii. Draw the symbol of the logic gate.

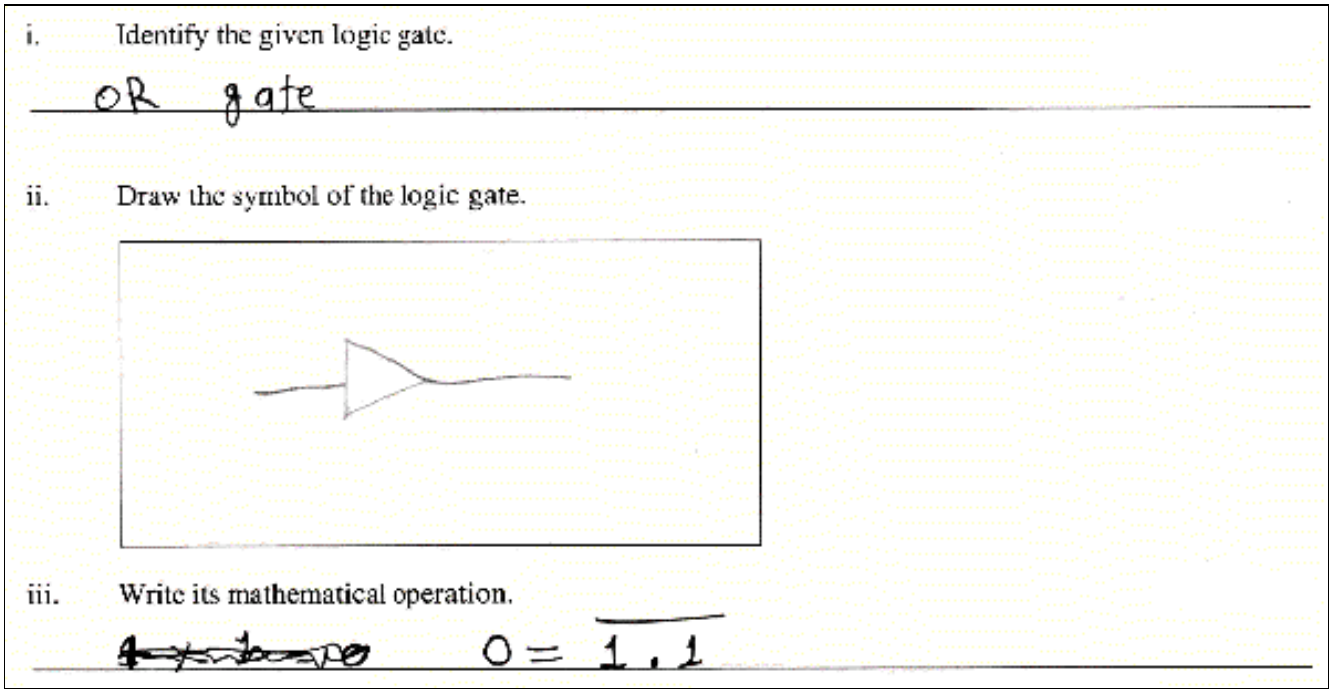


iii. Write its mathematical operation.

$X = \overline{A \cdot B}$ X = A AND B NOT. / So $X = \overline{A \cdot B}$

Weaker responses wrongly identified the logic gate, i.e. AND and OR gate. Some of the low scoring candidates failed to draw the correct symbol of NAND gate. Few of the candidates mentioned the binary system notation instead of the correct mathematical operation formula.

Example:



Question 8:

i. Complete the given statement:

Alpha (α) decay is the emission of helium _____.

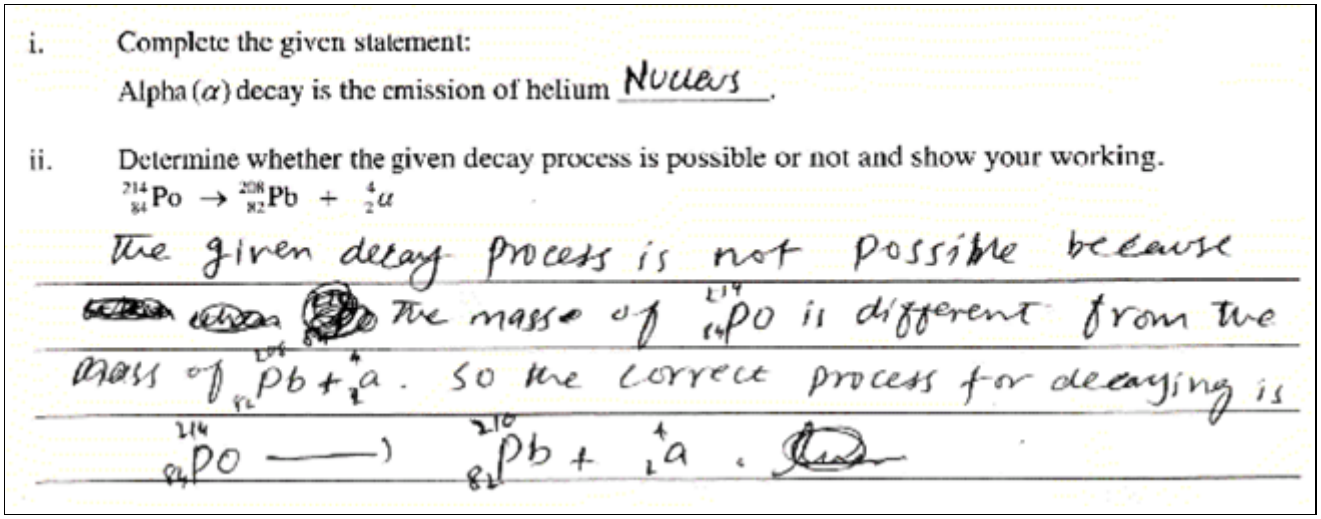
ii. Determine whether the given decay process is possible or not?

$^{214}_{84}\text{Po} \rightarrow {}^{208}_{82}\text{Pb} + {}^{4}_{2}\alpha$

Better responses correctly wrote 'nucleus' in part (i) and determined whether the given nuclear decay process will be possible or not by giving the following evidence in part (ii) of the question.

Charge: 84 = 82 + 2, Atomic mass number: 214 ≠ 208 + 4 and finally concluded that this nuclear decay process will not be possible.

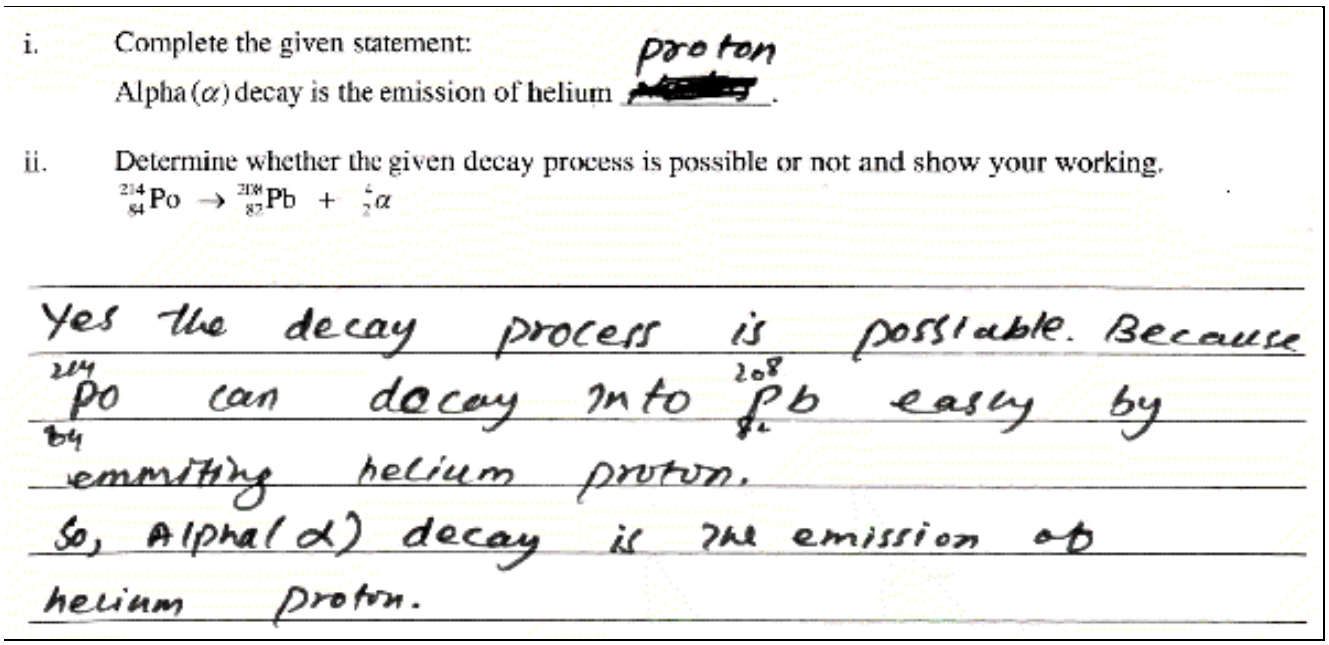
Example:


i. Complete the given statement:
Alpha (α) decay is the emission of helium Nucleus.

ii. Determine whether the given decay process is possible or not and show your working.
$^{214}_{84}Po \rightarrow {}^{208}_{82}Pb + {}^{4}_{2}u$
The given decay process is not possible because the masse of $^{214}_{84}Po$ is different from the mass of $^{208}_{82}Pb + {}^{4}_{2}a$. So the correct process for decaying is $^{214}_{84}Po \longrightarrow {}^{210}_{82}Pb + {}^{4}_{2}a$.

Weaker responses incorrectly identified that alpha decay is the emission of helium as proton and/ or electron. Some of the low scoring candidates erroneously determined that the nuclear decay process is possible.

Example:


i. Complete the given statement:
Alpha (α) decay is the emission of helium proton.

ii. Determine whether the given decay process is possible or not and show your working.
$^{214}_{84}Po \rightarrow {}^{208}_{82}Pb + {}^{4}_{2}\alpha$
Yes the decay process is possiable. Because $^{214}_{84}Po$ can decay into $^{208}_{82}Pb$ easly by emmiting helium proton.
So, Alphal (α) decay is the emission of helium proton.

Extended Response Questions (ERQs)
These questions offered a choice between part **a** and **b**

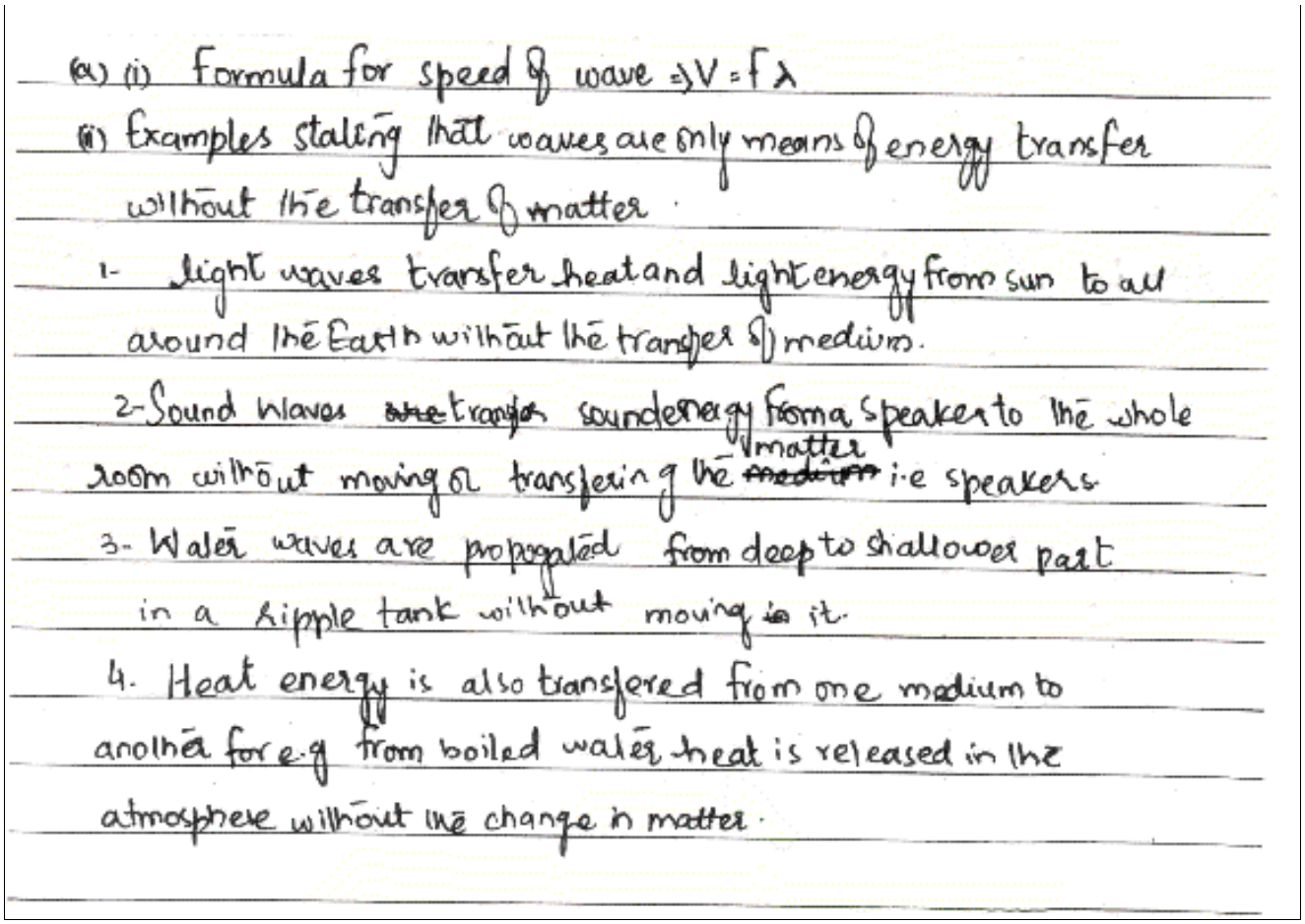
(**Note**: Majority of the candidates attempted Question **9b** and they scored well in this part.)

Question 9a:

i. Write formula for the speed of a wave.
'Waves are only the means of energy transfer, without the transfer of matter.'
Describe any FOUR examples in the light of the above statement.

Better responses correctly wrote the formula for the speed of wave, $V = f \lambda$. High scoring candidates correctly described the relevant examples of the energy transfer, without the transfer of matter, e.g. energy transferred without matter like sound energy from an instrument; light energy from the sun or sources like electric bulb, heat energy from the sun, energy transfer in a rope, waves produced in a slinky spring, in water and vibration of tunning forks.

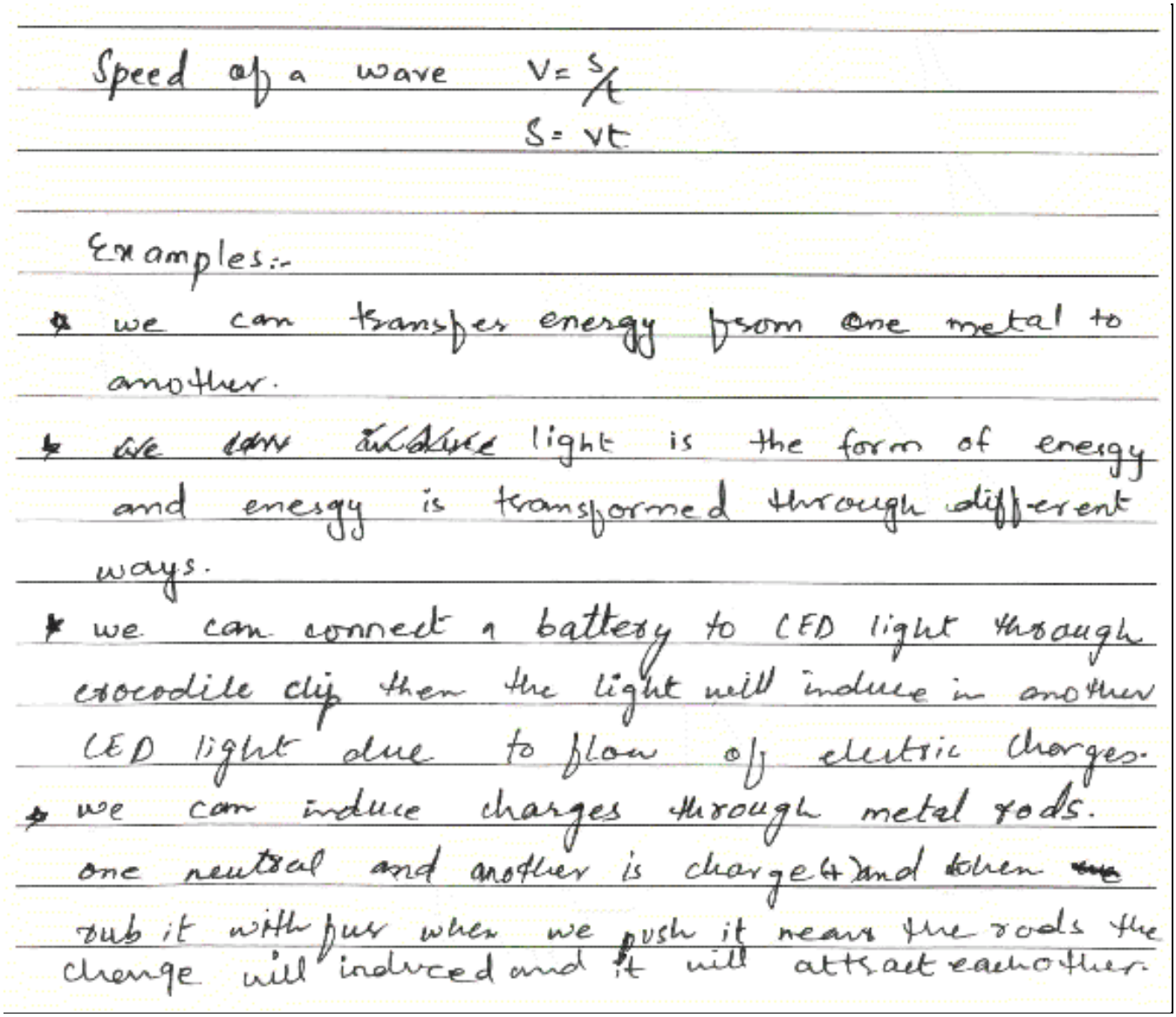
Example:


(a) (i) Formula for speed of wave => V = f λ
(ii) Examples stating that waves are only means of energy transfer without the transfer of matter.
1- Light waves transfer heat and light energy from sun to all around the Earth without the transfer of medium.
2- Sound waves ~~are~~ transfer sound energy from a speaker to the whole room without moving or transfering the ~~medium~~ matter i.e speakers.
3- Water waves are propogated from deep to shallower part in a ripple tank without moving in it.
4. Heat energy is also transfered from one medium to another for e.g from boiled water, heat is released in the atmosphere without the change in matter.

Weaker responses revealed that some candidates wrote the correct formula but were unable to describe the correct and relevant examples of energy transfer. Some of the low scoring candidates also failed to

write the correct formula, they mostly wrote S = v t; $S=\frac{v}{t}$. Such candidates also mentioned wrong examples, i.e. reflection, refraction, diffraction etc. Some of the candidates wrote about kinetic and potential energy, mechanical energy and electrical energy. Few of the candidates were confused about the uses and properties of waves.

Example:


Speed of a wave $V = S/t$

$S = vt$

Examples:-

* we can transfer energy from one metal to another.
* we can observe light is the form of energy and energy is transformed through different ways.
* we can connect a battery to LED light through crocodile clip then the light will induce in another LED light due to flow of electric charges.
* we can induce charges through metal rods. one neutral and another is charged and then rub it with fur when we push it nears the rods the charge will induced and it will attract eachother.

Question 9b

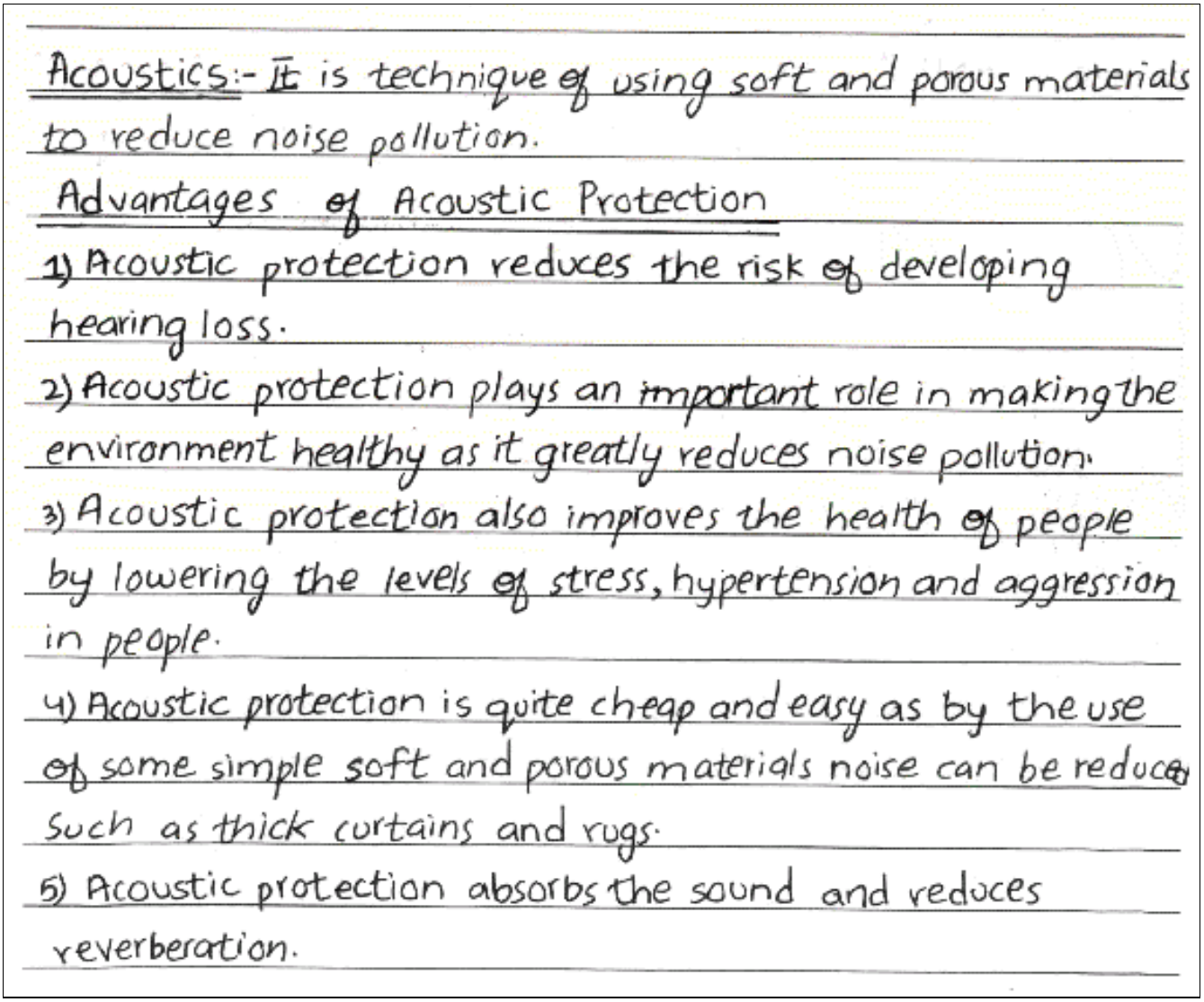
Zain is a social worker who wants to create awareness regarding the use of acoustic protection in the major cities of Pakistan. He is planning to publish a brochure to explain the advantages of acoustic protection to the general public.

As a physics student, suggest FIVE advantages of acoustic protection which should be published in this brochure.

Most candidates struggled to attempt this question. The understanding of acoustic protection was generally lacking.

Better responses suggested all five advantages of acoustic protection, e.g. help in reducing unpleasant echoes, minimises stress, protects hearing ability of individuals, reduces unwanted and unpleasant noise and helps in providing comfortable environment.

Example:



Acoustics:- It is technique of using soft and porous materials to reduce noise pollution.

Advantages of Acoustic Protection

1) Acoustic protection reduces the risk of developing hearing loss.

2) Acoustic protection plays an important role in making the environment healthy as it greatly reduces noise pollution.

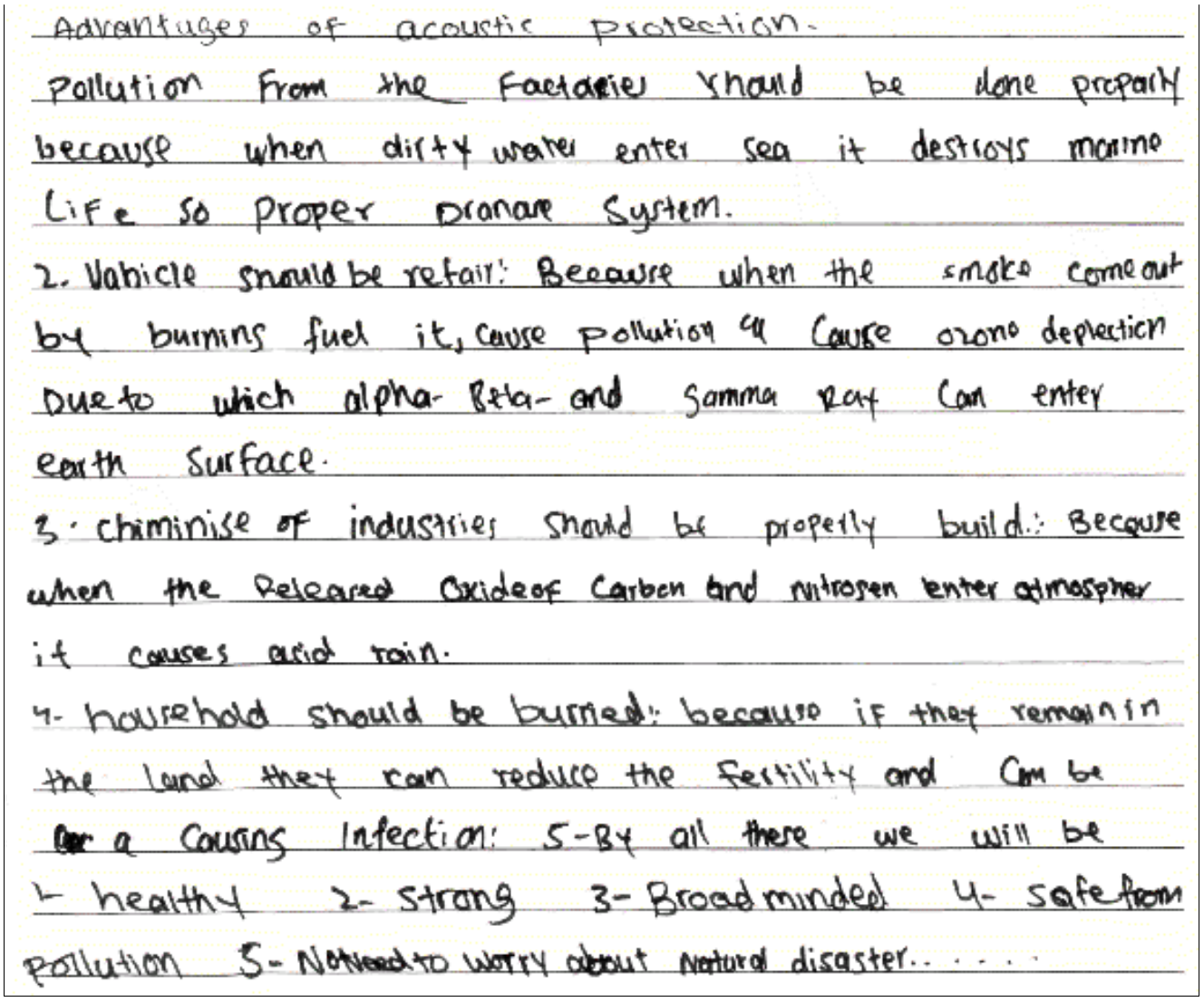
3) Acoustic protection also improves the health of people by lowering the levels of stress, hypertension and aggression in people.

4) Acoustic protection is quite cheap and easy as by the use of some simple soft and porous materials noise can be reduced such as thick curtains and rugs.

5) Acoustic protection absorbs the sound and reduces reverberation.

Weaker responses reflected that candidates lacked understanding of acoustic protection and its impact on environment. Most of the low scoring candidates wrote irrelevant and generic points, e.g. light and its production, pollution, causes of load shedding etc.

Example:



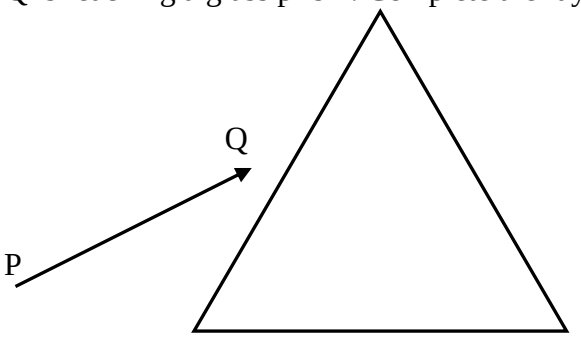
Advantages of acoustic protection.
Pollution from the Factories should be done properly because when dirty water enter sea it destroys marine Life so proper Dranage System.
2. Vehicle should be retair: Because when the smoke come out by burning fuel it, cause pollution & Cause ozone depletion Due to which alpha- Beta- and Gamma ray Can enter earth surface.
3. chiminise of industries should be propetly build.: Because when the Releared Oxide of Carbon and nitrogen enter atmospher it causes acid rain.
4- household should be burned: because if they remain in the land they can reduce the Fertility and Can be for a Causing Infection: 5-By all there we will be 1- healthy 2- Strong 3- Broad minded 4- safe from pollution 5- No need to worry about Natural disaster.....

(**Note**: The question offered a choice between part **a** and part **b**. Candidates chose to attempt both parts equally and scored well.)

Question 10a

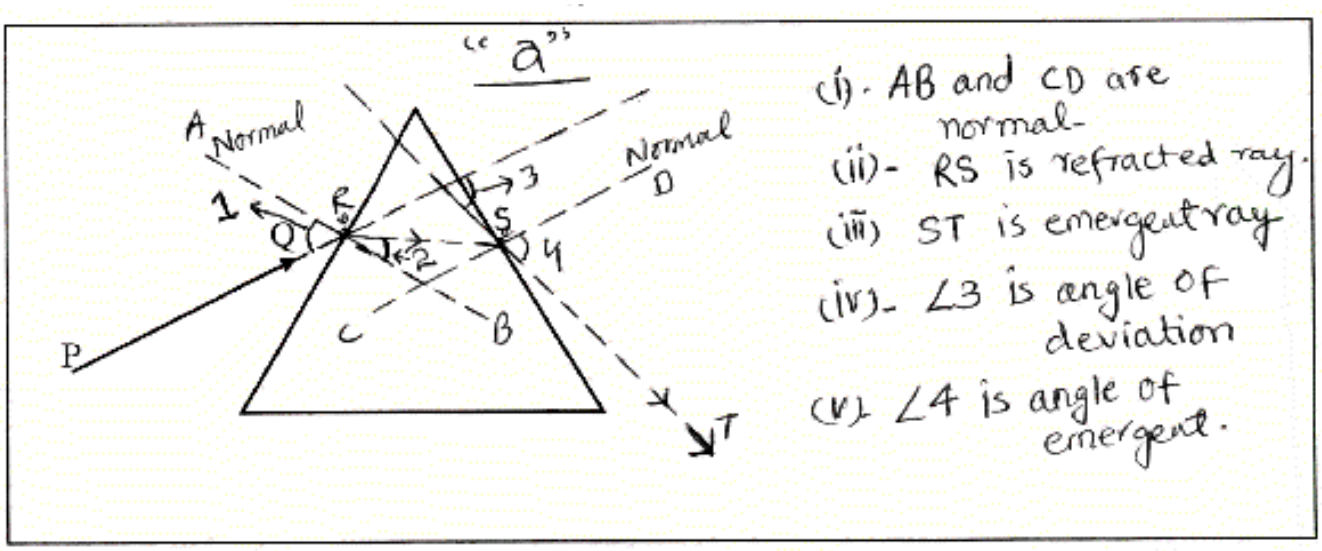
The given diagram shows an incident ray PQ is reaching a glass prism. Complete the ray diagram illustrating the following.

i. Normal
ii. Refracted ray
iii. Emergent ray
iv. Angle of deviation
v. Angle of emergent



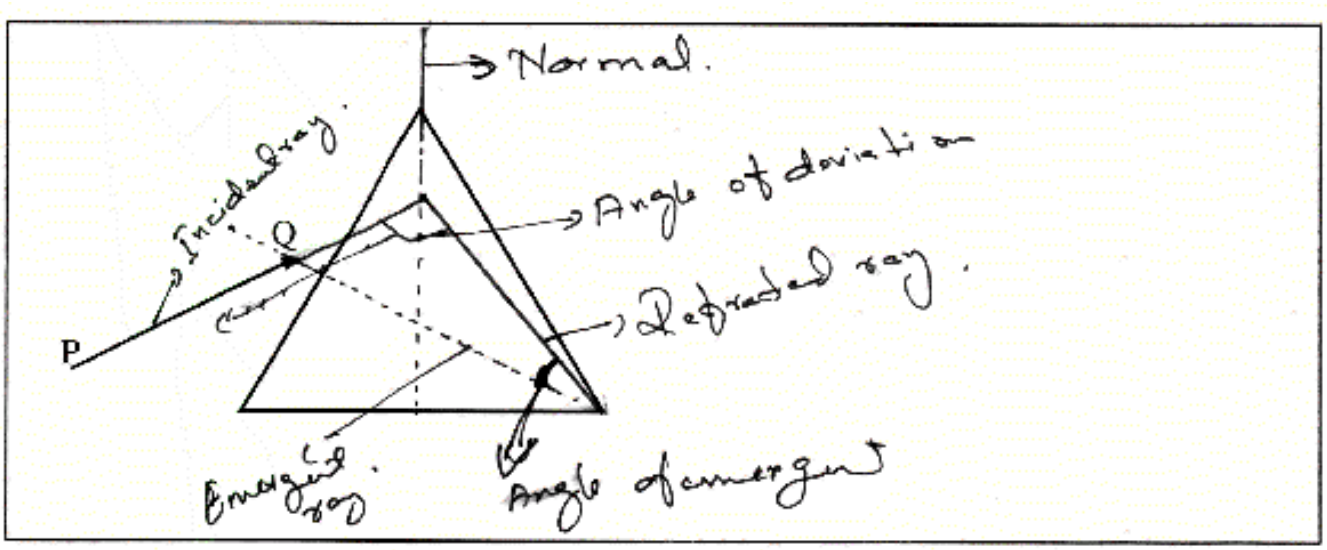
Better responses correctly completed the given ray diagram by showing the passage of light through a glass prism and illustrated all the required terms.

Example:



Weaker responses failed to complete the given ray diagram. Some of the low scoring candidates were confused in the labelling of normal, refracted ray and angle of deviation. Candidates mixed-up the concepts of refracted ray with emergent ray and angle of deviation with angle of emergent.

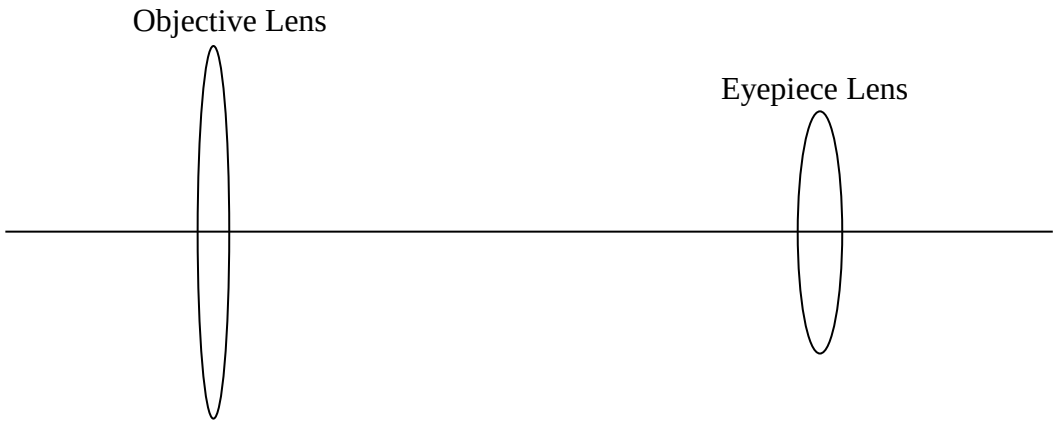
Example:



Question 10b

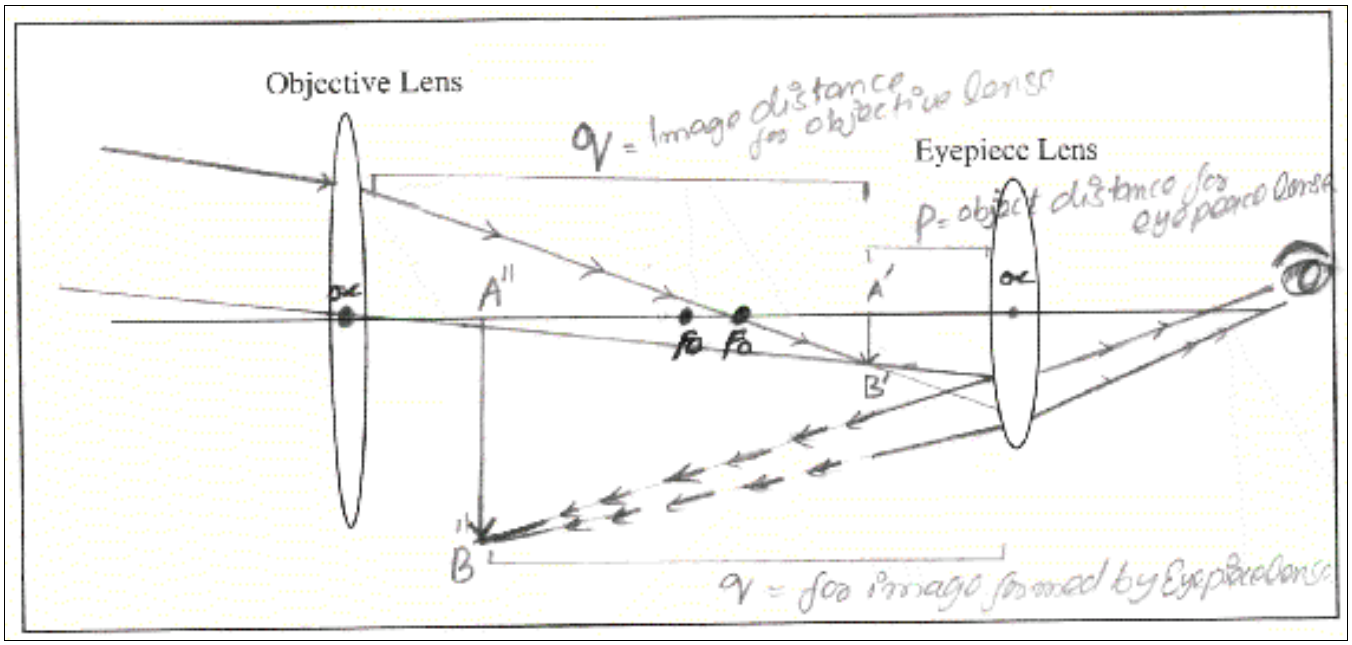
An astronomer uses a telescope to observe stars at night.

Complete the following ray diagram to show the refraction of light which allows him to see the stars through the telescope.



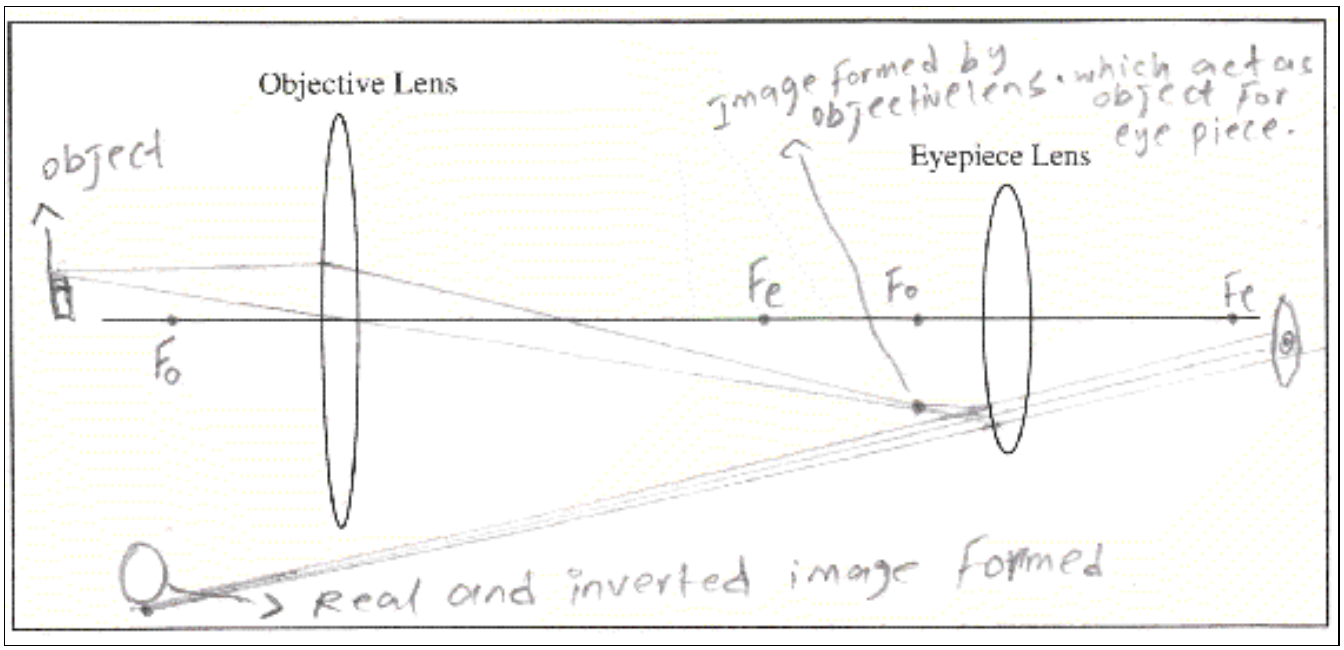
Better responses correctly completed the given ray diagram of telescope by showing the passage of light through glass lenses.

Example:



Weaker responses failed to complete the given ray diagram. Some of the low scoring candidates were confused in the labelling of position of object and image making correct direction of light rays.

Example:

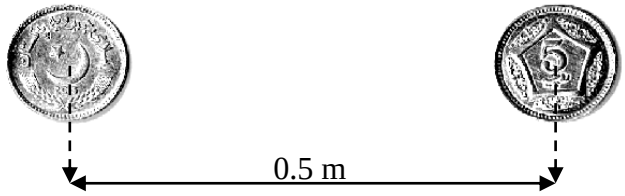


(**Note**: The question offered a choice between part **a** and part **b**. Majority of the candidates attempted Question **11a** and they scored well in this part.)

Question 11

Question 11a

The given two coins carry identical charge and experience a repulsive force of 5 N.



Calculate the value of charge on each coin.

(**Note**: The value of the constant of proportionality 'k' is 9×10^9 Nm^2/C^2)

Better responses displayed that candidates extracted data from the question, wrote the correct formula, $F = k\frac{q_1 \times q_2}{r^2}$, substituted values at the correct places in the formula, calculated the value of charge with S.I. unit, i.e. $q = 1.174 \times 10^{-5}\,C$.

Example:

Data: $F = 5N$

$k = 9 \times 10^9 Nm^2/C^2$

$q_1 q_2 = ?$

As we know that

$$F = k\frac{q_1 q_2}{r^2}$$

$$5N = \frac{9 \times 10^9 Nm^2C^{-2} q_1 q_2}{(0.5m)^2}$$

$$5N = \frac{9 \times 10^9 Nm^2C^{-2} q_1 q_2}{0.25 m^2}$$

$$1.25N = 9 \times 10^9 NC^{-2} q_1 q_2$$

$$q_1 q_2 = \frac{1.25N}{9 \times 10^9 NC^{-2}}$$

$$q_1 q_2 = 0.138 \times 10^{-9} C^2$$

$$q_1 q_2 = 1.38 \times 10^{-10} C^2$$

As q_1 and q_2 are identical so $q_1 = q_2$

The above equation becomes

$$q_1^2 = 1.38 \times 10^{-10} C^2$$

$$q_1 = \sqrt{1.38} \times \sqrt{10^{-10}} C$$

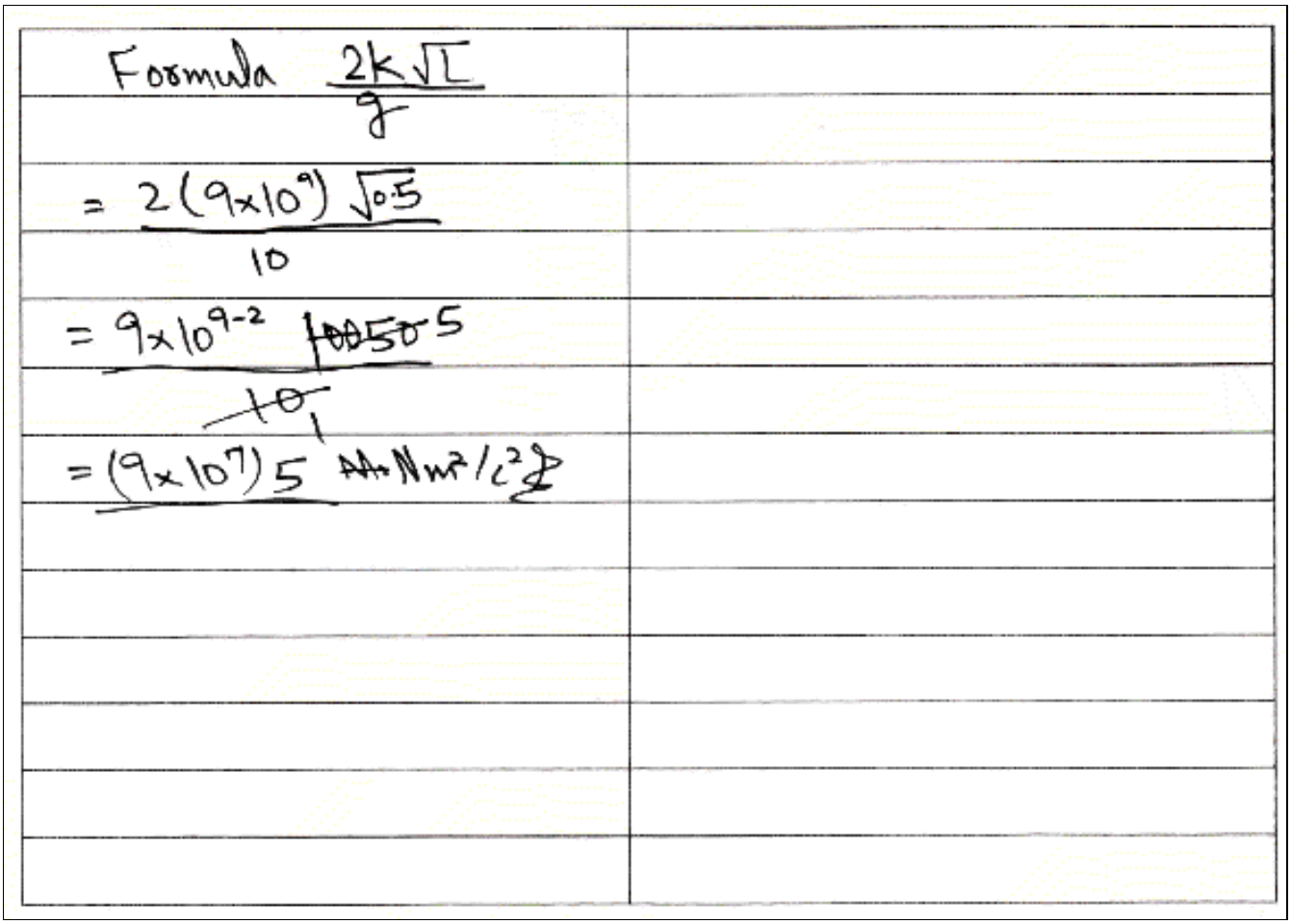
$$\boxed{q_1 = 1.17 \times 10^{-5} C}$$

As $q_1 = q_2$ so

$$q_2 = 1.17 \times 10^{-5} C$$

Weaker responses showed that candidates were unable to extract the data (values) from the question and incorrectly used the formula, e.g. $T = 2\pi\sqrt{\frac{l}{g}}$ and failed to obtain any single mark.

Example:



Question 11b

Assume a charged particle of 80 C is to be transferred through a potential difference of 40 V in time 20 s.

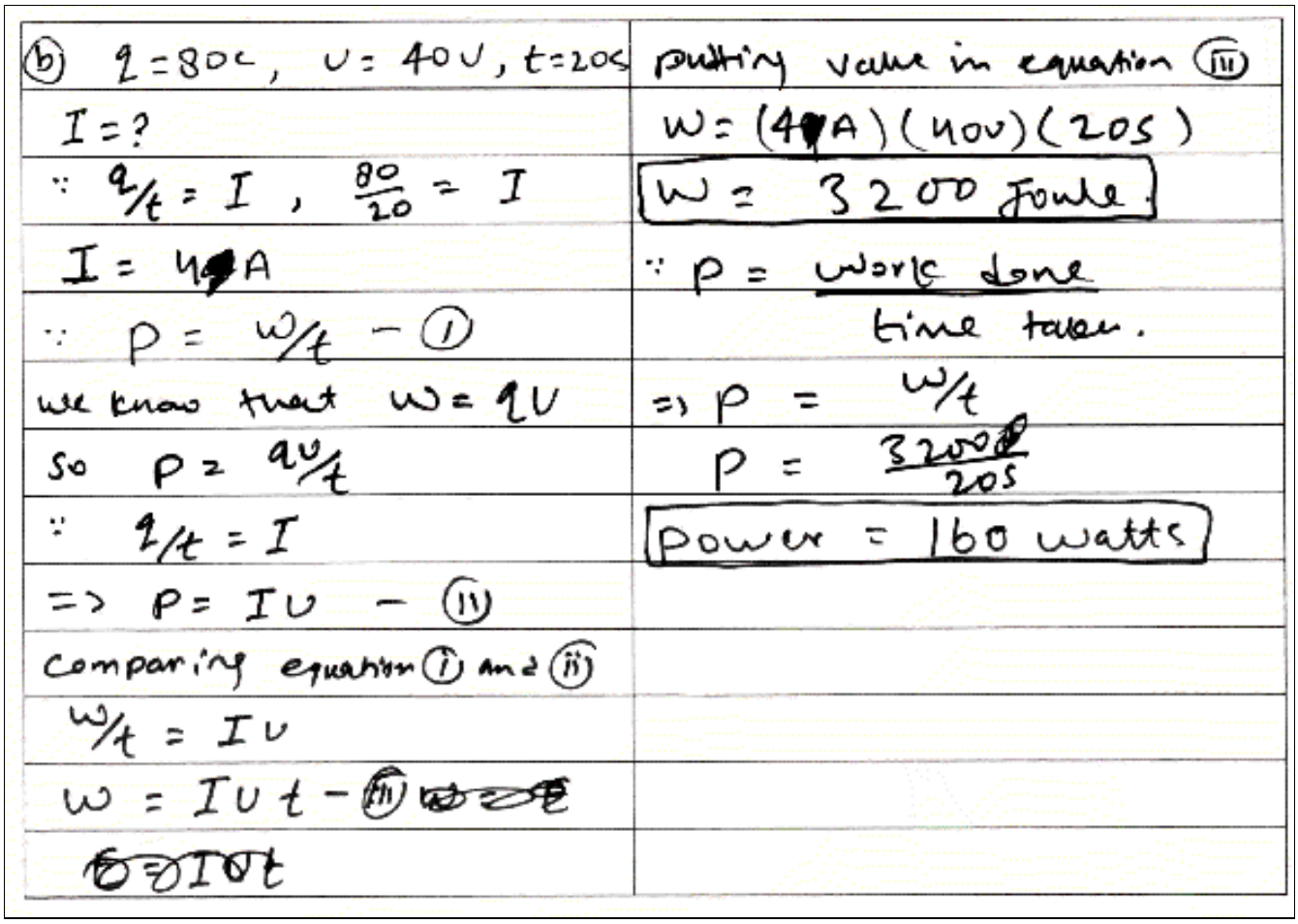
Calculate the following:

i. Work done

ii. Power

Better responses correctly extracted the data (values) from the question, wrote the correct formula of work done, $W = qV$ and formula of power, $P = \frac{W}{t}$ and calculated the correct values of work done, $W = 3200\,J$ and power, $P = 160\,W$. Some of the high scoring candidates used different formulae to solve this questions, they used $I = \frac{Q}{t}$; $W = V\,I\,t$; $V = I\,R$ and $W = I^2\,R\,t$, this indicated the good command of the candidates over the understanding of this topic.

Example:



Weaker responses reflected candidates' inability to extract data from the question. Most candidates were also unable to write the correct formula of work done and power. Low scoring candidates wrongly used the formulae, $W = \frac{CV}{S}$; $P = W\,C$; $P = \frac{V}{W}$ in their calculation and hence were unable to get any marks.

Example:

